

Table 1: Key tools and libraries that help with security in a Docker environment

Library / Tool	Usage	URL
Docker Bench for Security	Audit of Docker containers and benchmarking	https://github.com/docker/docker-bench-security
Cilium	Kernel layer security	https://github.com/cilium/cilium
Anchore	Inspection of container security with integration of policies	https://github.com/anchore/anchore-engine
OpenSCAP Workbench	Integration of security and privacy policies	https://www.open-scap.org/
Dagda	Scanning of viruses, malware, Trojans and other malicious traffic in Docker containers	https://github.com/eliasgranderubio/dagda
Notary	Integration of cryptography and deep level security	https://github.com/notaryproject/notary
Grafaes	Internal policies for privacy and security	https://grafeas.io/
Sysdig Falco	Deep level visibility and scan of containers with behavioural activity analysis	https://github.com/falcosecurity/falco
Banyanops Collector	Evaluation of Docker on different perspectives and dimensions	https://github.com/banyanops/collector
Dockscan	Security scanner against vulnerability	https://github.com/kost/dockscan
Batten	Docker audit toolkit	https://github.com/dockersecuritytools/batten
Inspec	Testing, audit and deep level analysis of the Docker environment	https://github.com/inspec/inspec

containers. A container's ability to access the system file directory might weaken the effectiveness of security measures. In rare cases, attackers who get root access to a container can use it to gain root access to the host too.

When it comes to container security, Docker has you covered at every stage: during development, during deployment, and during orchestration. However, the Docker daemon, container runtime, Dockerfile, and base images are all potential entry points for attackers.

So Docker does come with a few risks and vulnerabilities:

- Unsafe and unrestricted traffic with vulnerable communications
- Malicious and vulnerable container images
- Host system security loopholes
- Unrestricted access
- Exposure of Docker daemon socket

- Privacy leakages in Docker images
- Vulnerabilities with host kernel
- Container breakouts
- Kernel level threats

Free tools and libraries for Docker and containers

Though Docker is a secured environment, there is huge scope for research in its privacy, integrity and security aspects. Lots of libraries and frameworks are available and still under development to cope with the security leakages in a Docker environment (see Table 1). The Common Vulnerabilities and Exposures (CVE) database and the National Vulnerability Database (NVD) are

the key databases that receive a lot of attention from auditors and are used as standards by other organisations. After the container image is scanned, its contents are exposed, and comparisons are made against the vulnerability manifests in order to assess its security posture.

Organisations can benefit a lot from automating container audits and other container security activities since doing so allows teams to detect issues early in the build process. A number of free and open source libraries are still under development and will be released at continuous intervals to help businesses use them in real world scenarios. **END** 

 By: Dr Gaurav Kumar

The author is associated with various academic and research institutes for delivering expert lectures and conducting technical workshops on the latest technologies and tools.